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Hiring Committee
Institute of Biotechnology and Biochemical Engineering
TU Graz
Graz, Austria

To the Hiring Committee,

I am writing to express my profound interest in the PhD Position in Probabilistic Machine Learning for Biotechnology at TU Graz. As an AI Engineer with a rigorous background in mathematical modeling and real-world data uncertainty quantification, I am deeply motivated to collaborate on your research involving enzyme design and biotechnological optimization.

A core challenge in applying machine learning to complex biological processes—such as enzyme design—is managing uncertainty and missing data. In my "Cardiovascular Medical Digital Twin," I tackled this by designing a **Missingness-Aware Gate** that dynamically handles sensor dropouts in ICU telemetry data. This experience in building probabilistic and robust deep learning pipelines ensures that my models can reliably process the inherent noise and variability found in biotechnological datasets.

Furthermore, my engineering of a zero-retraining **PI_DeepONet** (Physics-Informed Operator Network) utilizing PyTorch Autograd highlights my capability to embed strict biological and physical constraints directly into neural architectures. I am eager to apply these probabilistic frameworks and my strong Python coding skills to accelerate the discovery and optimization of novel enzymes at TU Graz.

Enclosed are my CV and detailed project descriptions. I look forward to the possibility of discussing how my technical background aligns with your research goals.

Sincerely,

Ismail Aissaoui